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JEE Main Physics Practice Sheet — DPP-030

Topic: Kinetic Friction, Rolling Friction & Two-Block Friction Problems
Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A block of mass $m = 2$ kg is sliding along a rough horizontal floor with an initial speed of $v_0 = 10$ m/s. If the coefficient of kinetic friction is $\mu_k = 0.25$, the distance travelled by the block before it comes to rest is ($g = 10$ m/s^2):

(A) 10 m
(B) 20 m
(C) 40 m
(D) 5 m

Q2. Which of the following correctly describes the fundamental origin of rolling friction between a rolling wheel and the ground?

(A) Relative sliding velocity at the point of contact
(B) Local deformation of the contacting surfaces causing asymmetry in the normal reaction line
(C) Air drag acting on the top half of the wheel
(D) Electrostatic repulsion between smooth surfaces

Q3. A block slides down a rough inclined plane of inclination $\theta = 45°$ with constant velocity. The coefficient of kinetic friction μ_k between the block and the plane is:

(A) 0.5
(B) 1.0
(C) $\frac{1}{\sqrt{3}}$
(D) $\sqrt{3}$

Q4. A block takes twice as long to slide down a rough 45° inclined plane as it does to slide down an identical smooth 45° inclined plane. The coefficient of kinetic friction μ_k of the rough plane is:

(A) 0.25
(B) 0.50
(C) 0.75
(D) 0.80

Q5. A block of mass $m_1 = 2$ kg is placed on top of another block of mass $m_2 = 3$ kg. The coefficient of static friction between the blocks is $\mu_s = 0.4$, and the floor beneath m_2 is frictionless. The maximum horizontal force F that can be applied to the lower block m_2 so that both blocks move together without slipping is ($g = 10$ m/s^2):

(A) 20 N
(B) 8 N
(C) 12 N
(D) 25 N

Q6. In Question 5, if the horizontal force applied to the lower block m_2 is $F = 30$ N, the acceleration of the upper block m_1 ($\mu_k = 0.3$) and the lower block m_2 are respectively:

(A) 3 m/s^2 and 8 m/s^2
(B) 4 m/s^2 and 6 m/s^2
(C) 6 m/s^2 and 6 m/s^2
(D) 2 m/s^2 and 10 m/s^2

Q7. A block of mass $m_1 = 4$ kg rests on a slab of mass $m_2 = 6$ kg on a smooth floor. The coefficient of friction between m_1 and m_2 is $\mu_s = 0.5$. If a horizontal force $F = 12$ N is applied to the upper block m_1, the frictional force acting on the slab m_2 is ($g = 10$ m/s^2):

(A) 20 N
(B) 7.2 N
(C) 12 N
(D) 4.8 N

Q8. Which of the following relations correctly ranks the magnitude of friction coefficients under identical contact surface conditions?

(A) $\mu_{\text{rolling}} > \mu_k > \mu_s$
(B) $\mu_s > \mu_k > \mu_{\text{rolling}}$
(C) $\mu_k > \mu_s > \mu_{\text{rolling}}$
(D) $\mu_s = \mu_k = \mu_{\text{rolling}}$

Q9. A conveyor belt moves horizontally with a constant speed $v = 4$ m/s. A box is gently placed on the belt with zero initial speed. If $\mu_k = 0.2$, the distance travelled by the box relative to the ground before it stops slipping on the belt is ($g = 10$ m/s^2):

(A) 2 m
(B) 4 m
(C) 8 m
(D) 1 m

Q10. In Question 9, the distance travelled by the box relative to the belt during the slipping phase is:

(A) 2 m
(B) 4 m
(C) 8 m
(D) 1 m

Q11. A block of mass m is projected with speed v_0 up a rough inclined plane of angle θ ($\mu_k < \tan\theta$). The ratio of the time of ascent t_{up} to the time of descent t_{down} back to the starting point is:

(A) $\sqrt{\frac{\sin\theta - \mu_k\cos\theta}{\sin\theta + \mu_k\cos\theta}}$

(B) $\sqrt{\dfrac{\sin\theta + \mu_k \cos\theta}{\sin\theta - \mu_k \cos\theta}}$

(C) $\dfrac{\sin\theta - \mu_k \cos\theta}{\sin\theta + \mu_k \cos\theta}$

(D) 1

Q12. A block of mass $M = 10$ kg is placed on a rough horizontal surface with $\mu_k = 0.2$. A horizontal force $F(t) = 4t$ N (where t is in seconds) is applied starting from $t = 0$. The speed of the block at $t = 10$ s is ($g = 10$ m/s^2, assume $\mu_s = \mu_k$):

(A) 5 m/s

(B) 10 m/s

(C) 15 m/s

(D) 20 m/s

Q13. A heavy block of mass m is pulled with a constant horizontal force $F > \mu_k mg$ across a rough horizontal floor. The rate of heat energy dissipated by kinetic friction as the block moves with speed v is:

(A) Fv

(B) $\mu_k mgv$

(C) $\frac{1}{2}\mu_k mgv^2$

(D) $(F - \mu_k mg)v$

Q14. A small coin rests on a horizontal turntable at a distance $r = 20$ cm from the axis of rotation. The coefficient of static friction between the coin and the turntable is $\mu_s = 0.5$. The maximum angular speed ω of the turntable at which the coin does not slip is ($g = 10$ m/s^2):

(A) 2.5 rad/s

(B) 5.0 rad/s

(C) 10.0 rad/s

(D) 1.58 rad/s

Q15. A chain of length L and mass M is held on a rough horizontal table such that a length y hangs over the edge. If the coefficient of static friction between the chain and the table is μ_s, the maximum fraction y/L of the chain that can hang over without sliding is:

(A) $\dfrac{\mu_s}{1+\mu_s}$

(B) $\dfrac{\mu_s}{1-\mu_s}$

(C) $\dfrac{1}{1+\mu_s}$

(D) μ_s

Q16. A plank of mass M is placed on a smooth horizontal floor. A block of mass m is projected with speed u on the top surface of the plank (μ_k between block and plank). The common final velocity of the block and the plank when relative slipping ceases is:

(A) $\dfrac{mu}{M+m}$

(B) $\dfrac{Mu}{M+m}$

(C) $\dfrac{u}{2}$

(D) $\dfrac{(M-m)u}{M+m}$

Q17. In Question 16, the total mechanical energy converted into heat due to kinetic friction between the block and plank during the process is:

(A) $\frac{1}{2}mu^2$

(B) $\frac{1}{2}\left(\dfrac{Mm}{M+m}\right)u^2$

(C) $\frac{1}{2}\left(\dfrac{M^2}{M+m}\right)u^2$

(D) $\mu_k mgL$

Q18. A block of mass $m = 2$ kg is kept on a rough inclined plane of angle 30° ($\mu_s = 0.6, \mu_k = 0.5$). A force F is applied parallel to the incline upward. The minimum value of F to prevent the block from sliding down is ($g = 10$ m/s^2, $\sqrt{3} \approx 1.73$):

(A) 0 N

(B) 10 N

(C) $10 - 6\sqrt{3} \approx -0.38$ N $\implies$ 0 N

(D) 20 N

Q19. A solid cylinder of mass M and radius R rolls without slipping down an inclined plane of angle θ. The magnitude and direction of the static frictional force acting on the cylinder is:

(A) $\frac{1}{3}Mg\sin\theta$, directed up the incline

(B) $\frac{1}{2}Mg\sin\theta$, directed down the incline

(C) $\mu_s Mg\cos\theta$, directed up the incline

(D) Zero

Q20. A car travels at speed v on a flat horizontal road. The coefficient of static friction between the tyres and the road is μ_s. The minimum stopping distance without skidding (using maximum static friction) is:

(A) $\dfrac{v^2}{\mu_s g}$

(B) $\dfrac{v^2}{2\mu_s g}$

(C) $\dfrac{2v^2}{\mu_s g}$

(D) $\dfrac{v}{\mu_s g}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-030)

Kinetic Friction, Rolling Friction & Problems on Friction

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	A	11	A	16	A
2	B	7	B	12	A	17	B
3	B	8	B	13	B	18	A
4	C	9	B	14	B	19	A
5	A	10	B	15	A	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (B)
Kinetic friction opposes sliding:

$$f_k = \mu_k mg = 0.25 \times 2 \times 10 = 5 \text{ N}$$

Deceleration of the block:

$$a = \frac{f_k}{m} = \mu_k g = 0.25 \times 10 = 2.5 \text{ m/s}^2$$

Using $v^2 = u^2 - 2as$ with final velocity $v = 0$:

$$s = \frac{v_0^2}{2a} = \frac{10^2}{2 \times 2.5} = \frac{100}{5} = 20 \text{ m}$$

Sol 2. Option (B)
When a body rolls, both the body and the contact surface undergo microscopic elastic/inelastic deformation. As the wheel rolls forward, the normal reaction shifts slightly ahead of the geometric center line, creating a counter-torque that opposes rolling. Hence rolling friction arises from surface deformation.

Sol 3. Option (B)
For a block sliding down an incline with constant speed (zero acceleration):

$$\sum F_{\parallel} = 0 \implies mg\sin\theta - f_k = 0$$

$$mg\sin\theta = \mu_k mg\cos\theta \implies \mu_k = \tan\theta = \tan 45^\circ = 1.0$$

Sol 4. Option (C)
Acceleration down a smooth 45° incline: $a_{\text{smooth}} = g\sin 45^\circ = \frac{g}{\sqrt{2}}$.
Acceleration down a rough 45° incline: $a_{\text{rough}} = g(\sin 45^\circ - \mu_k \cos 45^\circ) = \frac{g}{\sqrt{2}}(1 - \mu_k)$.
Since distance $s = \frac{1}{2}at^2$ is the same:

$$t \propto \frac{1}{\sqrt{a}} \implies \frac{t_{\text{rough}}}{t_{\text{smooth}}} = \sqrt{\frac{a_{\text{smooth}}}{a_{\text{rough}}}} = 2$$

$$\frac{a_{\text{smooth}}}{a_{\text{rough}}} = 4 \implies \frac{1}{1-\mu_k} = 4 \implies 1-\mu_k = 0.25 \implies \mu_k = 0.75$$

Sol 5. Option (A)
The only force accelerating the upper block m_1 forward is static friction from m_2:

$$f_{s1}^{\max} = \mu_s m_1 g = 0.4 \times 2 \times 10 = 8 \text{ N}$$

Maximum common acceleration without slipping:

$$a_{\max} = \frac{f_{s1}^{\max}}{m_1} = \mu_s g = 0.4 \times 10 = 4 \text{ m/s}^2$$

Applying Newton's second law to the combined system $(m_1 + m_2)$:

$$F_{\max} = (m_1 + m_2)a_{\max} = (2+3) \times 4 = 20 \text{ N}$$

Sol 6. Option (A)
Since applied force $F = 30 \text{ N} > F_{\max} = 20 \text{ N}$, slipping occurs between the blocks.
Kinetic friction operates between the blocks:

$$f_k = \mu_k m_1 g = 0.3 \times 2 \times 10 = 6 \text{ N}$$

Acceleration of upper block m_1:

$$a_1 = \frac{f_k}{m_1} = \frac{6}{2} = 3 \text{ m/s}^2$$

Acceleration of lower block m_2:

$$a_2 = \frac{F - f_k}{m_2} = \frac{30-6}{3} = \frac{24}{3} = 8 \text{ m/s}^2$$

Sol 7. Option (B)
Limiting friction between m_1 and m_2: $f_s^{\max} = \mu_s m_1 g = 0.5 \times 4 \times 10 = 20 \text{ N}$.
Max acceleration for no slipping (accelerated by slab's friction): $a_{\max} = \frac{f_s^{\max}}{m_2} = \frac{20}{6} = 3.33 \text{ m/s}^2$.
Max force on m_1 for combined motion: $F_{\max} = (m_1 + m_2)a_{\max} = 10 \times \frac{20}{6} = 33.3 \text{ N}$.
Since applied $F = 12 \text{ N} < 33.3 \text{ N}$, both blocks move together:

$$a_{\text{common}} = \frac{F}{m_1 + m_2} = \frac{12}{4+6} = 1.2 \text{ m/s}^2$$

The friction force driving the slab m_2 is:

$$f = m_2 a_{\text{common}} = 6 \times 1.2 = 7.2 \text{ N}$$

Sol 8. Option (B)
For given materials, the interlocking of asperities makes static friction coefficient highest. Once motion begins, asperities shear and slide with less average interlocking ($\mu_k < \mu_s$). In rolling, the relative motion is ideally zero and resistance arises purely from tiny surface deformation, making μ_{rolling} drastically smaller ($\approx 10^{-2}$ to 10^{-3}). Hence $\mu_s > \mu_k > \mu_{\text{rolling}}$.

Sol 9. Option (B)
Acceleration of the box due to kinetic friction:

$$a = \mu_k g = 0.2 \times 10 = 2 \text{ m/s}^2$$

Time to reach belt speed $v = 4$ m/s:

$$t = \frac{v}{a} = \frac{4}{2} = 2 \text{ s}$$

Distance travelled by box relative to the ground:

$$s_{\text{ground}} = \frac{v^2}{2a} = \frac{4^2}{2 \times 2} = \frac{16}{4} = 4 \text{ m}$$

Sol 10. Option (B)
Distance travelled by the belt during slipping time $t = 2$ s:

$$s_{\text{belt}} = v_{\text{belt}} \cdot t = 4 \times 2 = 8 \text{ m}$$

Relative displacement of box with respect to the belt:

$$s_{\text{rel}} = s_{\text{belt}} - s_{\text{ground}} = 8 \text{ m} - 4 \text{ m} = 4 \text{ m}$$

Sol 11. Option (A)
Deceleration during ascent: $a_{\text{up}} = g(\sin\theta + \mu_k \cos\theta)$.
Acceleration during descent: $a_{\text{down}} = g(\sin\theta - \mu_k \cos\theta)$.
Since distance $s = \frac{1}{2}at^2 \implies t = \sqrt{\frac{2s}{a}}$:

$$\frac{t_{\text{up}}}{t_{\text{down}}} = \sqrt{\frac{a_{\text{down}}}{a_{\text{up}}}} = \sqrt{\frac{\sin\theta - \mu_k \cos\theta}{\sin\theta + \mu_k \cos\theta}}$$

Sol 12. Option (A)
Frictional force: $f = \mu_k Mg = 0.2 \times 10 \times 10 = 20$ N.
The block starts moving when $F(t) = f \implies 4t = 20 \implies t_0 = 5$ s.
For $t \geq 5$ s, acceleration is:

$$a(t) = \frac{F(t) - f}{M} = \frac{4t - 20}{10} = 0.4t - 2$$

Integrating velocity from $t = 5$ to $t = 10$ s:

$$v(10) = \int_5^{10} (0.4t - 2)\, dt = \left[0.2t^2 - 2t\right]_5^{10}$$

$v(10) = (0.2(100) - 20) - (0.2(25) - 10) = 0 - (-5) = 5$ m/s

Sol 13. Option (B)
Rate of heat dissipation equals the power dissipated by the non-conservative kinetic friction force:

$$P_{\text{friction}} = f_k \cdot v = (\mu_k mg)v$$

Sol 14. Option (B)
Static friction provides the required centripetal acceleration:

$$m\omega^2 r \leq \mu_s mg \implies \omega \leq \sqrt{\frac{\mu_s g}{r}}$$

$$\omega_{\text{max}} = \sqrt{\frac{0.5 \times 10}{0.2}} = \sqrt{\frac{5}{0.2}} = \sqrt{25} = 5 \text{ rad/s}$$

Sol 15. Option (A)
Weight of hanging part: $W_{\text{hang}} = \left(\frac{M}{L}y\right) g$.
Normal reaction on table: $N = \left(\frac{M}{L}(L - y)\right) g$.
Limiting friction: $f_s^{\max} = \mu_s N = \mu_s \frac{M}{L}(L - y)g$.
For equilibrium:

$$\frac{M}{L}yg = \mu_s \frac{M}{L}(L - y)g \implies y = \mu_s(L - y)$$

$$y(1 + \mu_s) = \mu_s L \implies \frac{y}{L} = \frac{\mu_s}{1 + \mu_s}$$

Sol 16. Option (A)
No external horizontal force acts on the (block + plank) system. Linear momentum is conserved:

$$mu = (M + m)v_{\text{common}} \implies v_{\text{common}} = \frac{mu}{M + m}$$

Sol 17. Option (B)
Initial kinetic energy: $K_i = \frac{1}{2}mu^2$.
Final kinetic energy: $K_f = \frac{1}{2}(M + m)v_{\text{common}}^2 = \frac{1}{2}(M + m)\left(\frac{mu}{M+m}\right)^2 = \frac{1}{2}\frac{m^2u^2}{M+m}$.
Energy dissipated as heat:

$$\Delta E = K_i - K_f = \frac{1}{2}mu^2\left(1 - \frac{m}{M + m}\right) = \frac{1}{2}\left(\frac{Mm}{M + m}\right)u^2$$

Sol 18. Option (A)
Downhill gravity component: $mg \sin 30° = 20 \times 0.5 = 10$ N.
Maximum static friction up the plane:

$$f_s^{\max} = \mu_s mg \cos 30° = 0.6 \times 20 \times \frac{\sqrt{3}}{2} = 6\sqrt{3} \approx 10.39 \text{ N}$$

Since $f_s^{\max} > mg \sin 30°$, the block does not slide even without any external force ($F = 0$ N).

Sol 19. Option (A)
For a rolling solid cylinder ($I = \frac{1}{2}MR^2$): Equations of motion: 1. $Mg\sin\theta - f_s = Ma$ 2. $f_s R = I\alpha = \left(\frac{1}{2}MR^2\right)\left(\frac{a}{R}\right) \implies f_s = \frac{1}{2}Ma$
Substituting into (1):

$$Mg\sin\theta - \frac{1}{2}Ma = Ma \implies a = \frac{2}{3}g\sin\theta$$

$$f_s = \frac{1}{2}M\left(\frac{2}{3}g\sin\theta\right) = \frac{1}{3}Mg\sin\theta \quad \text{(directed up the incline)}$$

Sol 20. Option (B)
Without skidding, static friction between tyre and ground provides the maximum braking deceleration:

$$a_{\text{max}} = \mu_s g$$

Using $v^2 = 2as$:

$$s_{\text{min}} = \frac{v^2}{2a_{\text{max}}} = \frac{v^2}{2\mu_s g}$$